

IGNAT'YEVO, Russian Federation

WMO: 313289

Lat: 50.425N

Lon: 127.412E

Elev: 195

StdP: 99.01

Time Zone: 9.00 (E09)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-30.1	-36.0	0.1	-31.8	-33.9	0.2	-29.5	8.2	-17.9	7.3	-18.1	1.2	250	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.2	21.0	28.9	20.4	27.1	19.8	23.6	27.5	22.6	26.3	21.7	25.2	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.3	25.3	21.2	16.2	24.2	20.2	15.3	23.3	71.1	27.5	67.5	26.0	64.3	25.6	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.7	6.7	-36.6	33.3	1.8	1.4	-37.9	34.2	-38.9	35.1	-39.9	35.8	-41.2	36.8		
			-36.4	25.2	1.9	1.0	-37.8	25.9	-38.8	26.5	-39.9	27.1	-41.2	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-21.1	-17.0	-6.7	4.5	12.4	18.1	21.6	19.6	12.7	2.5	-10.4	-19.9
	DBStd	15.96	4.79	5.67	6.63	5.20	3.88	3.39	2.48	2.94	4.45	4.99	6.89	4.62
	HDD10.0	4234	963	756	519	181	18	1	0	0	23	236	612	927
	HDD18.3	6373	1221	989	777	417	187	46	4	19	176	490	862	1185
	CDD10.0	1114	0	0	0	14	93	243	358	297	103	4	0	0
	CDD18.3	211	0	0	0	1	3	38	104	59	6	0	0	0
	CDH23.3	1829	0	0	0	23	112	453	751	426	65	0	0	0
	CDH26.7	450	0	0	0	6	24	143	190	78	8	0	0	0
Wind	WSAvg	2.8	2.1	2.3	3.1	3.9	3.6	2.7	2.6	2.5	3.0	3.1	2.7	2.3
Precipitation	PrecAvg	548	6	6	9	24	53	84	146	106	66	26	12	10
	PrecMax	791	30	30	29	90	127	214	322	246	134	86	26	31
	PrecMin	360	0	0	0	0	9	10	13	25	13	3	1	3
	PrecStd	111	6	6	9	19	32	49	70	55	37	20	7	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.2	1.2	12.2	26.2	29.8	32.8	32.1	30.9	27.9	20.2	10.9	-6.2
		MCWB	-10.1	-2.9	5.2	12.0	15.0	19.8	23.2	23.1	18.8	10.6	5.1	-7.7
	2%	DB	-10.9	-2.1	8.8	20.9	26.8	30.2	30.2	28.8	25.1	16.8	6.2	-8.8
		MCWB	-12.3	-4.9	3.1	9.8	14.2	20.0	22.5	21.4	16.4	9.2	2.5	-9.7
	5%	DB	-12.2	-5.0	6.0	16.9	24.0	28.1	28.9	27.1	22.9	13.2	2.8	-10.9
		MCWB	-13.3	-6.7	1.0	7.8	12.8	18.7	21.9	20.6	16.2	7.4	0.1	-11.8
	10%	DB	-14.1	-7.9	3.8	14.0	21.2	25.9	27.1	25.9	20.2	10.9	-0.8	-13.1
		MCWB	-15.0	-9.3	-0.3	5.9	12.0	18.2	20.9	19.9	14.4	5.6	-2.2	-13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.1	-2.4	5.1	13.1	17.3	22.6	24.7	25.0	20.4	13.3	4.9	-7.6
		MCDB	-8.4	1.0	11.7	25.1	25.4	27.5	28.8	29.0	23.2	17.8	8.6	-6.6
	2%	WB	-12.2	-4.7	3.1	10.3	15.6	21.4	23.9	23.3	18.6	10.2	2.9	-9.6
		MCDB	-10.9	-2.5	7.9	18.9	22.3	26.8	27.9	26.7	21.6	13.7	7.6	-8.7
	5%	WB	-13.6	-6.7	1.1	8.3	14.4	20.3	23.1	22.1	17.2	8.1	-0.1	-11.7
		MCDB	-12.6	-5.1	5.5	15.7	21.4	25.8	27.2	25.2	20.9	11.9	2.6	-10.8
	10%	WB	-15.0	-9.3	-0.4	6.4	13.1	19.3	22.3	21.0	16.0	6.1	-2.2	-13.9
		MCDB	-14.3	-8.1	3.4	12.8	19.4	24.4	25.9	24.5	19.6	10.2	-0.5	-13.2
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.8	11.9	12.7	13.2	11.9	10.4	10.8	12.1	11.0	9.2	8.6
		MCDBR	9.3	12.3	12.6	17.2	18.0	16.2	12.3	12.7	14.7	15.1	11.8	9.8
		MCWBR	8.1	10.0	8.0	9.5	7.9	7.3	5.9	6.8	7.3	9.2	8.4	8.7
	5% WB	MCDBR	9.1	11.8	12.4	15.5	14.6	12.8	10.1	10.5	11.5	12.8	10.9	9.4
		MCWBR	8.0	9.7	8.1	8.9	7.6	6.8	5.8	6.3	6.4	8.9	8.3	8.4
Clear-Sky Solar Irradiance	taub		0.269	0.300	0.372	0.509	0.568	0.498	0.479	0.427	0.390	0.383	0.315	0.276
	taud		2.165	2.095	1.954	1.782	1.726	2.012	2.118	2.252	2.274	2.128	2.155	2.120
	Ebn at Noon		751	817	807	729	706	770	778	805	794	704	672	668
	Edh at Noon		75	106	148	199	220	167	148	123	108	103	77	68
All-Sky Solar Radiation	RadAvg		1.66	2.86	4.41	5.01	5.44	5.87	5.43	4.82	3.86	2.56	1.65	1.29
	RadStd		0.06	0.10	0.28	0.53	0.45	0.74	0.45	0.33	0.33	0.15	0.09	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page